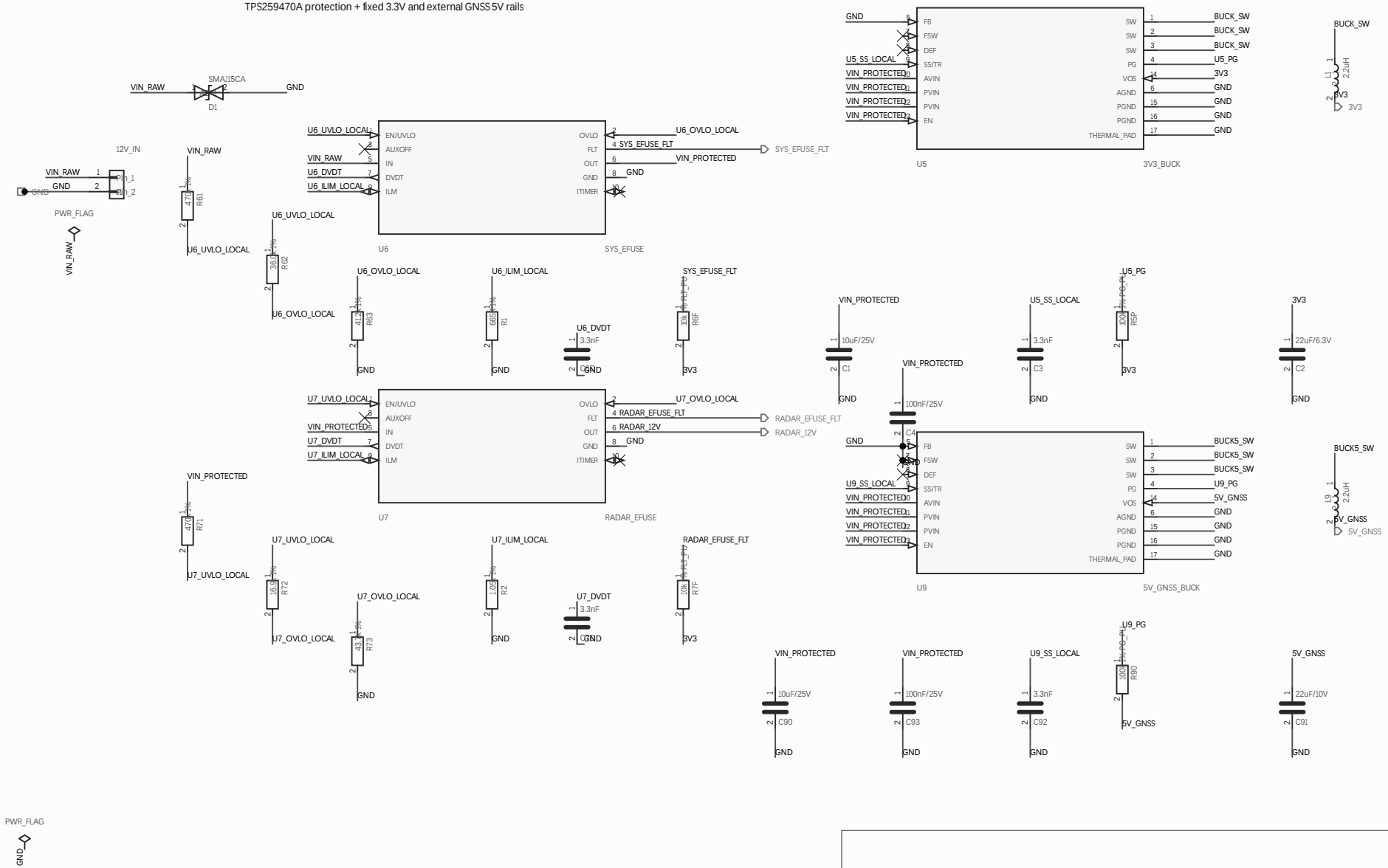
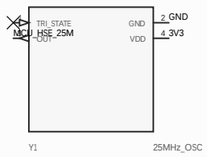
POWER_INPUT — Rev A authoritative capture
TPS259470A protection + fixed 3.3V and external GNSS 5V rails



Sheet: /	
File: 10_POWER_INPUT.kicad_sch	
Title: EVT_SYS_CARRIER	
Size: A4	Date:
KiCad E.D.A. kicad (5.1.4)-1	Rev:
	Id: 1/1

MCU_CORE — STM32H753IIT6
P0 time capture pins remain fixed by PINMAP_SYS_001.csv















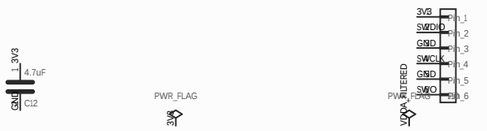






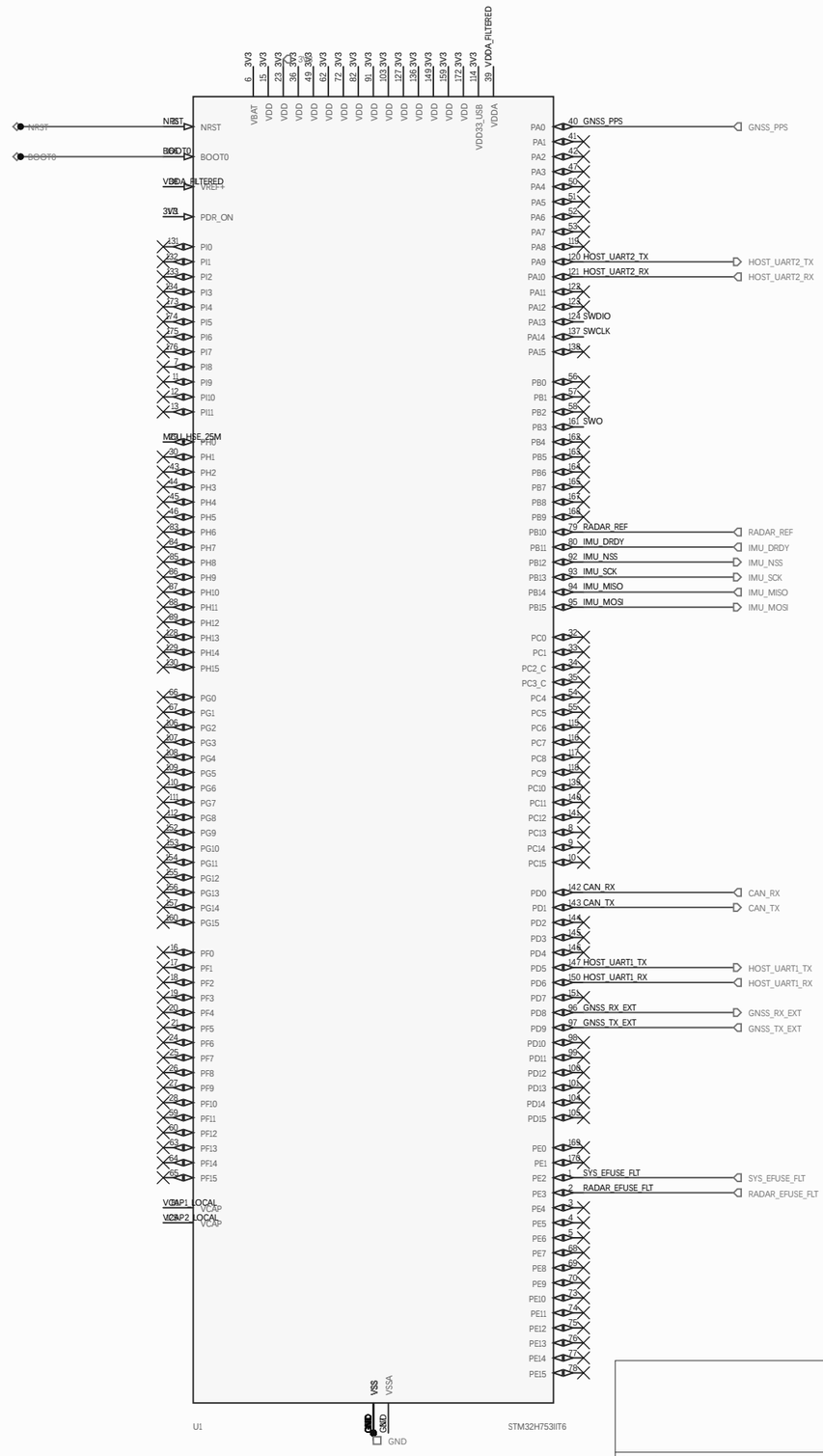










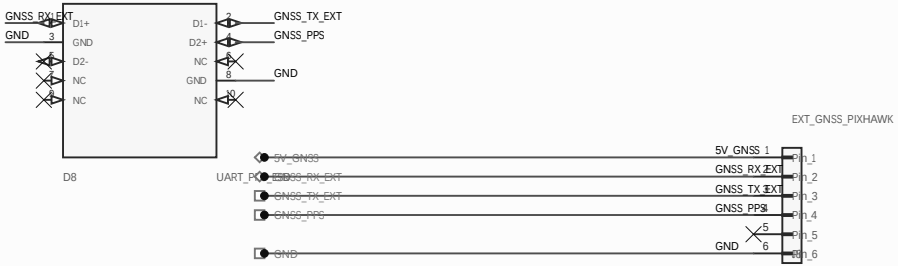


Sheet: /
File: 20_MCU_CORE.kicad_sch

Title: EVT_SYS_CARRIER

Size: A3	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

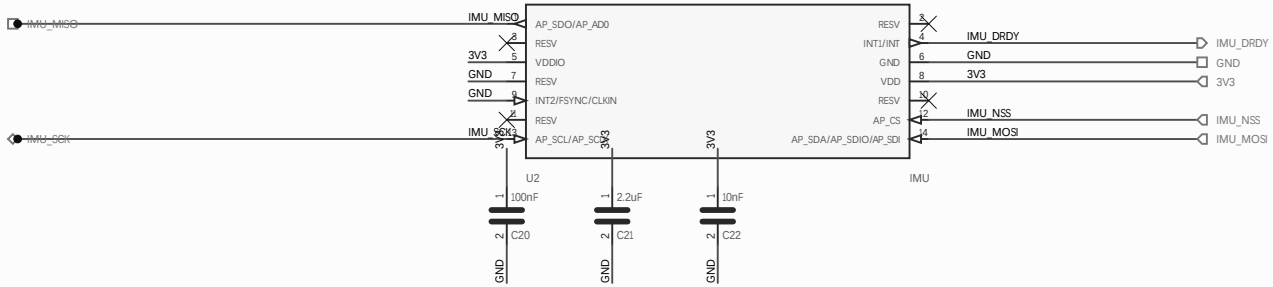
TIME_GNSS — external simpleRTK3B Pro / mosaic-X5 interface
Optional untrusted GNSS aiding: 5V, USART3 and 3.3V PPS



J8 Pixhawk adapter: 1 5V, 2 receiver RX,
3 receiver TX, 4 PPS, 5 event NC, 6 GND.
D8 protects UART/PPS. The receiver may be absent
or quarantined without stopping carrier operation.

Sheet: /		
File: 30_TIME_GNSS.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A5	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

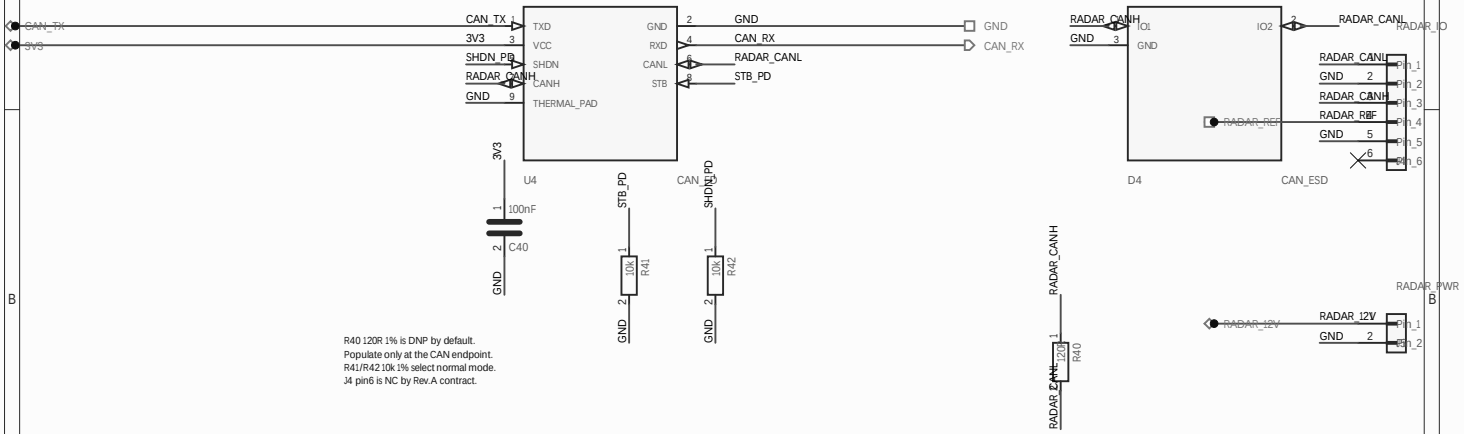
IMU — ICM-42688-P SPI2 + hardware DRDY
P0 IMU_DRDY -> STM32 PB11/TIM2_CH4



C20/C21: VDD decoupling. C22: 10nF VDDIO.
Pin7 RESV and pin9 FSYNC connect to GND
per the Rev.A pin contract.

Sheet: /		
File: 40_IMU.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A5	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

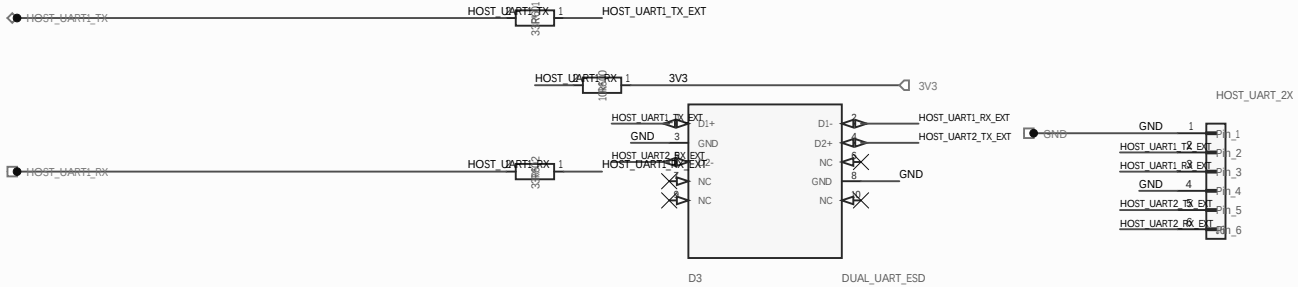
RADAR_IO_CAN — external AWR2944-class EVM interface
CAN-FD control/metadata + P0 RADAR_REF timestamp + protected 12V



R40 120R 1% is DNP by default.
Populate only at the CAN endpoint.
R41/R42 10k 1% select normal mode.
J4 pin6 is NC by Rev.A contract.

Sheet: /		
File: 50_RADAR_IO_CAN.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A5	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

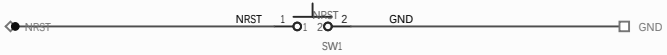
HOST_UART — two independent 3.3V UART interfaces
USART2 on PD5/PD6 + USART1 on PA9/PA10; TX/RX/GND only

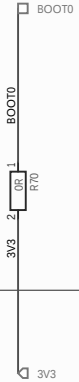


J6: 1 GND, 2 U1 TX, 3 U1 RX,
4 GND, 5 U2 TX, 6 U2 RX.
R601-R604: 33R series. R600/R605: 10k RX pull-ups.
D3 protects all four signal lines.
No supply voltage is exported.

Sheet: /		
File: 60_HOST_UART.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A5	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

DEBUG_BOOT — SWD, reset and boot service access
No vehicle actuation/control interfaces





Primary Tag-Connect SWD remains on MCU_CORE.
SW1 pulls NRST low. R70 is a DNP
service override only.

Sheet: /
File: 70_DEBUG_BOOT.kicad_sch

Title: EVT_SYS_CARRIER

Size: A5	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

EVT System Carrier

External radar front-end: TI AWR2944PEVM

Carrier contains no 76–81 GHz RF routing

Sheet: POWER_INPUT

3V3
5V_GNSS
RADAR_12V
SYS_EFUSE_FLT
RADAR_EFUSE_FLT
GND

File: POWER_INPUT.kicad_sch.sch

3V3
5V_GNSS
RADAR_12V
SYS_EFUSE_FLT
RADAR_EFUSE_FLT
GND

Sheet: MCU_CORE

NRST
BOOT0
RADAR_EFUSE_FLT
SYS_EFUSE_FLT
HOST_UART2_RX
HOST_UART1_RX
CAN_RX
GNSS_TX_EXT
IMU_MISO
IMU_DRDY
RADAR_REF
GNSS_PPS
3V3
GND

IMU_NSS
IMU_SCK
IMU_MOSI
GNSS_RX_EXT
CAN_TX
HOST_UART1_TX
HOST_UART2_TX

File: MCU_CORE.kicad_sch.sch

NRST
BOOT0
RADAR_EFUSE_FLT
SYS_EFUSE_FLT
HOST_UART2_RX
HOST_UART1_RX
CAN_RX
GNSS_TX_EXT
IMU_MISO
IMU_DRDY
RADAR_REF
GNSS_PPS
3V3
GND

Sheet: TIME_GNSS

GNSS_PPS
GNSS_TX_EXT

GNSS_RX_EXT
5V_GNSS
GND

File: TIME_GNSS.kicad_sch.sch

GNSS_PPS
GNSS_TX_EXT

GNSS_RX_EXT
5V_GNSS
GND

Sheet: IMU

IMU_DRDY
IMU_MISO

IMU_MOSI
IMU_SCK
IMU_NSS
3V3
GND

File: IMU.kicad_sch.sch

IMU_DRDY
IMU_MISO

IMU_MOSI
IMU_SCK
IMU_NSS
3V3
GND

Sheet: RADAR_IO_CAN

CAN_RX
RADAR_REF

RADAR_12V
CAN_TX
3V3
GND

File: RADAR_IO_CAN.kicad_sch.sch

CAN_RX
RADAR_REF

RADAR_12V
CAN_TX
3V3
GND

Sheet: HOST_UART

HOST_UART1_RX
HOST_UART2_RX

HOST_UART2_TX
HOST_UART1_TX
3V3
GND

File: HOST_UART.kicad_sch.sch

HOST_UART1_RX
HOST_UART2_RX

HOST_UART2_TX
HOST_UART1_TX
3V3
GND

Sheet: DEBUG_BOOT

NRST

3V3
GND
BOOT0

File: DEBUG_BOOT.kicad_sch.sch

NRST
GND

NRST
BOOT0
GND

Sheet: /		
File: EVT_SYS_CARRIER.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A2	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1